

Institutional Blueprint: Signal Intake, BLEI-E Risk Engine Integration
& Hedge Desk Orchestration

1. Purpose & System Scope

The **Interest Developer.jpm** module serves as the primary operational and analytical bridge connecting the **BLEI-E Factor Engine** and **NIL Valuation Data Fabric** directly to the **Hedge Desk** and **Quant R&D; Lab**. The core mission of the system is to turn raw market interest (signals, biometric anomalies, structural dislocations, and quantitative hypotheses) into rigorously backtested, governed, and tradeable overlay strategies.

Hedge Desk Users:	Portfolio Managers (PMs), Risk Officers (CRO), Execution Traders, Structuring Desk.
Quant R&D; Users:	Quantitative Researchers (QR), Data Scientists, Infrastructure Engineers.
Core Function:	Systematically ingest signals → generate hypotheses → backtest → validate risk → deploy overlays.

2. High-Level Architecture & Layer Decomposition

Layer ID	Module Name	Primary Operational Function	Core Data Entities / APIs
L1	Signal Intake Layer	Ingests biometric, virality, and market data with physiological range checks	/api/factors/blei-e
L2	Interest Taxonomy	Registers ideas via standardized interest_ticket entities across 5 classes	interest_ticket (P1-P4)
L3	Research Pipeline	Orchestrates automated hypothesis testing, model fitting, and backtesting	Hypothesis → Backtest Engine
L4	Hedge Desk Layer	Maps candidate strategies to tradeable execution objects & risk limits	strategy_candidate, strategy_live
L5	Governance & Audit	Enforces immutable audit trails, approval sign-offs, and RBAC controls	WORM logs, audit_event
L6	Feedback Loop	Performs Weekly Interest Reviews, real P&L; tracking, and edge decay audit	Weekly Review Dashboard

3. Step-by-Step Design: Signal to Hedge Action

Step 1 & 2: Interest Taxonomy & Ticket Registry

Class Code	Interest Type Name	Description & Domain Scope
PERF	Performance Interest	New BLEI-E factor behavior, athlete biometric signals, physiological patterns.
SPONSOR	Sponsor Interest	Sponsor-athlete correlation shifts, contagion paths, brand equity dislocations.
STRUCTURE	Market Structure Interest	Spread behavior, basis dislocations, implied volatility surface dynamics.
PRODUCT	Product Interest	NIL-linked structured products, customized overlays, note structures.

RISK	Risk Interest	Failure mode identification, extreme tail event monitoring, systemic contagion.
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Data Entity: interest_ticket

interest_ticket Data Entity Schema:

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{ id: UUID, created_by: String, desk: 'hedge'|'r&d'|'joint', interest_type: 'PERF'|'SPONSOR'|'STRUCTURE'|'PRODUCT'|'RISK', description: Text, linked_signals: [BLEI_E, NIL, SPONSOR], priority: 'P1'..'P4', status: 'new'|'in_research'|'candidate'|'live'|'archived' }
```

Step 3 & 4: BLEI-E Data Integration & R&D; Research Pipeline

The system connects R&D; directly to the **BLEI-E Data Fabric** via dedicated read-only endpoints: /api/factors/blei-e, /api/nil/market, and /api/sponsor/risk. For every registered ticket, the **Research Pipeline Orchestrator** executes a 6-stage standardized lifecycle:

Stage	Pipeline Stage	Execution & Analytical Logic	Output Deliverable
1	Hypothesis Definition	Formulate targeted hypothesis (e.g., 'SEF-E spikes predict sponsor vol 30-90 min ahead')	Formal Hypothesis Spec
2	Data Slicing	Extract precise time window, asset universes, factor deltas, and control variables	Clean Feature Matrix
3	Model Choice	Apply regression, classification, time-series, or regime-detection algorithms	Fitted Quantitative Model
4	Backtesting	Simulate P&L; hit rate, maximum drawdown, Sharpe ratio, and turnover metrics	Backtest Performance Suite
5	Risk Review	Stress testing under extreme regimes, liquidity checks, tail risk analysis	Risk Profile Matrix
6	Outcome Classification	Classify ticket based on empirical thresholds into: reject, monitor, candidate_strategy	Decision Directive

Step 5, 6 & 7: Hedge Desk Integration, Governance & Review

When an R&D; ticket yields a candidate_strategy, it is mapped into Hedge Desk production objects (strategy_id, signal_source, asset_universe, entry/exit rules, risk_limits, capacity). Access is exposed via the Strategy Catalog UI, Risk Hooks, Simulation Mode (paper trading), and limited live deployment.

Governance Rule: No direct P&L; impact is permitted without full backtest validation, formal risk sign-off, and capacity review. Strict isolation is maintained between the **R&D; Sandbox** and **Hedge Desk Production**. A mandatory **Weekly Interest Review** evaluates top tickets, candidate strategies, retired ideas, and risk incidents.

4. BLEI-E Risk Engine Integration & 6-Layer Risk Fabric

The **Interest Developer.jpm** system embeds the 6-layer **BLEI-E Risk Management Framework**, which formalizes the **Dashawn Bledsoe Principle**:

The Dashawn Bledsoe Principle

“Trade sponsors and sports stocks using biometric-linked risk signals. Treat physiology as market data. Treat virality as volatility. Treat fan-base energy as revenue. Treat stress as contagion.”

Layer	Name	Technical & Quantitative Logic	Integrated Operational Control
L1	Signal Ingestion	Ingests biometric, virality, and market data; physiological range validation	Prevents invalid sensor data injection
L2	Factor Normalization	Calculates Z-scores, percentile ranks, and vol scaling across 5 BLEI-E deltas	Normalizes PMF-E, SEF-E, VVE-E, FBR-E, CSE-E

L3	Cross-Asset Mapping	OLS beta estimation with Newey-West HAC robust standard errors	Removes autocorrelation & heteroskedasticity
L4	CRS Risk-State Engine	Aggregates multi-layer inputs into a unified 0–4 CRS risk score	Triggers automatic risk overlay states
L5	Overlay Engine	Generates targeted hedges across sponsor, media, betting & franchise equities	Outputs real-time hedge execution orders
L6	Governance & Audit	Enforces WORM audit logs, HIPAA/GDPR privacy, RBAC, SEC alternative-data rules	Ensures compliance & factor lineage tracking

5. Technical Entities, Schemas & System Microservices

Core Entity	Key Fields / Attributes	System Role & Lifecycle Context
interest_ticket	id, created_by, desk, interest_type, description, linked_signals, priority, status	Tracks initial curiosity/hypothesis from creation to archiving.
strategy_candidate	candidate_id, ticket_id, signal_source, asset_universe, backtest_sharpe, max_dd	Container for validated R&D; models awaiting risk sign-off.
strategy_live	strategy_id, candidate_id, allocated_capital, current_pnl, live_sharpe, risk_limits	Production strategy executing on Hedge Desk with real P&L.;
risk_profile	profile_id, strategy_id, crs_state, var_99, stress_beta, liquidity_buffer	Dynamic risk parameter set enforced by risk engine hooks.
audit_event	event_id, timestamp, actor_id, entity_type, action, prior_state, new_state	Immutable WORM audit log entry for compliance tracking.

- Core Infrastructure Microservices:**
- interest-registry-service: Manages ticket state transitions, metadata, and data handles.
 - research-orchestrator-service: Executes automated feature generation, model training, and backtesting.
 - hedge-strategy-service: Interfaces with Hedge Desk execution engines and order management systems (OMS).
 - risk-governance-service: Calculates CRS risk states, enforces risk limits, and writes WORM audit records.
 - data-fabric-gateway: Exposes high-throughput read-only APIs for BLEI-E factor deltas and NIL valuations.

6. Pod-Level Developer & Production Deployment Workflow

Production execution is managed by a dedicated 5-role **Quant Pod: Portfolio Manager (PM), Quant Researcher (QR), Data Engineer (DE), Chief Risk Officer (CRO), and Compliance Officer**. The end-to-end developer deployment flow follows an 8-stage institutional lifecycle:

Step	Workflow Phase	Action Description & Milestone Requirement
1	Log an Interest	Create interest_ticket specifying class (PERF, SPONSOR, STRUCTURE, PRODUCT, RISK).
2	Attach Data	Link BLEI-E factor handles, NIL valuations, and sponsor stock tickers to ticket.
3	Run R&D; Pipeline	Execute hypothesis, data slicing, quantitative model fitting, and multi-factor backtest.
4	Promote Candidate	If backtest Sharpe, hit rate, and drawdown pass thresholds, promote to candidate_strategy.
5	Register Desk	Register candidate with Hedge Desk catalog UI, set risk limits, and attach risk hooks.
6	Deploy Sandbox	Initiate paper trading or limited live deployment with restricted notional allocation.
7	Monitor & Iterate	Track live P&L; and execution metrics vs. backtest to detect edge decay or regime shifts.

8	Promote or Retire	Transition strategy to full live production deployment or archive if edge decays.
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Econometric Validation Standards

Statistical Validation Gateways: All candidate models must pass rigorous econometric verification before live deployment: **GRS (Gibbons-Ross-Shanken) Tests** for factor efficiency, **Fama-MacBeth Regressions** for risk premium estimation, and **Newey-West HAC Corrections** to account for serial correlation and heteroskedasticity in financial time series.